

Defining reference values for body composition indices by magnetic resonance imaging in UK Biobank

Liang Dai¹ , Xiao-yan Huang^{1,2}, Yue-qi Lu³, Yu-yang Liu^{3,4}, Cong-ying Song¹, Jing-wen Zhang², Jing Li^{1,2}, Yue Zhang², Ying Shan^{1*} & Yu Shi^{5*}

¹Clinical Research Academy, Peking University Shenzhen Hospital, Peking University, Shenzhen, Guangdong, China; ²Renal Division, Peking University Shenzhen Hospital, Peking University, Shenzhen, China; ³BGI-Shenzhen, Shenzhen, China; ⁴College of Life Sciences, University of Chinese Academy of Sciences, Beijing, China; ⁵Department of Ultrasound, Peking University Shenzhen Hospital, Peking University, Shenzhen, Guangdong, China

Abstract

Background Magnetic resonance imaging (MRI) is the gold standard for evaluating body composition. However, the reference ranges have not been established.

Methods Three lean tissue and seven adipose tissue parameters based on MRI data from the UK Biobank were used in this study. Participants with European ancestry and data on at least one parameter were screened. Age- and sex-specific percentile curves were generated using the lambda–mu–sigma method. Three levels of reference ranges were provided, which were equivalent to the mean $\pm$ 1 standard deviation (SD), 2 SDs and 2.5 SDs.

Results The final analysis set for each parameter ranged from 4842 to 14 148 participants (53.4%–56.6% women) with a median age of 61. For lean tissue parameters, compared with those at age 45, the median total lean tissue volume and total thigh fat-free muscle volume at age 70 were 2.83 and 1.73 L, and 3.02 and 1.51 L lower in men and women, respectively. The median weight-to-muscle ratios at age 45 were 0.51 and 0.83 kg/L lower compared with those at age 70 in men and women, respectively. Adipose tissue parameters showed inconsistent differences. In men, the median muscle fat infiltration, visceral adipose tissue (VAT) volume, total abdominal adipose tissue index and abdominal fat ratio were 1.48%, 0.32 L, 0.08 L/m² and 0.4 higher, and the median abdominal subcutaneous adipose tissue (ASAT) volume and total adipose tissue volume were 0.47 and 0.41 L lower, respectively, at age 70 than at age 45. The median total trunk fat volume was approximately 9.53 L at all ages. In women, the median muscle fat infiltration and VAT volume were 1.68% and 0.76 L higher, respectively, at age 70 than at age 45. The median ASAT volume, total adipose tissue volume, total trunk fat volume, total abdominal adipose tissue index and abdominal fat ratio were 0.35 L, 0.78 L, 1.12 L, 0.49 L/m² and 0.06 higher, respectively, at age 60 than at age 45. The medians of the former three parameters were 0.33 L, 0.14 L and 0.20 L lower, at age 70 than at age 60. The medians of the latter two parameters were approximately 3.64 L/m² and 0.55 at ages between 60 and 70.

Conclusions We have established reference ranges for MRI-measured body composition parameters in a large community-dwelling population. These findings provide a more accurate assessment of abnormal adipose and muscle conditions.

Keywords Abdominal body composition; Lambda–mu–sigma method; Magnetic resonance imaging; Reference ranges

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*Correspondence to: Ying Shan, Clinical Research Academy, Peking University Shenzhen Hospital, Peking University, 1120 Lianhua Road, Futian District, Shenzhen 518036, Guangdong, China. Email: shanying@pkusz.com;

Yu Shi, Department of Ultrasound, Peking University Shenzhen Hospital, Peking University, 1120 Lianhua Road, Futian District, Shenzhen 518036, Guangdong, China. Email: shiyu@pkusz.com

Liang Dai and Xiao-yan Huang contributed equally to this study.

Introduction

Lean and fat mass play important roles that affect health-related outcomes. It has been illustrated that a U-shaped relationship exists between lean body mass and all-cause mortality.¹ Additionally, the relationship between adipose tissue and cardio-metabolic health has been repeatedly proven.^{2,3} Hence, body composition evaluations benefit both clinicians and patients in terms of understanding the potential risks of negative health events. Body mass index (BMI), which is the classical body measurement index, has continuously been criticized because of its poor accuracy in assessing specific body compositions.^{4,5} Meanwhile, waist circumference (WC) and waist-hip ratio, which serve as more sensitive indicators of visceral obesity, cannot comprehensively interpret potential disease risks.^{6,7} Therefore, more comprehensive evaluations of body composition are required in clinical practice.

Bioelectrical impedance analysis (BIA) and dual-energy X-ray absorptiometry (DXA) are the two techniques commonly used for body composition analysis. BIA can estimate adipose and muscle mass based on whole-body electrical conductivity.⁸ Owing to its convenience and low cost, BIA is frequently used for body composition screening. As the name suggests, DXA calculates body tissue mass based on absorption and depletion differences of X-rays with two energies.⁹ As a snapshot technique, DXA can provide compositional indices for specific limbs. However, both instruments have their own limitations, such as inconsistent results from different brands, interference from body hydration status and the inability to assess intra-muscular fat, which would lead to the obesity paradox or muscle quality overestimation.^{10–13, S1–S2} Hence, more accurate instruments are required for body composition evaluation.

The magnetic resonance imaging (MRI) technique utilizes the interaction of hydrogen nuclei with the magnetic field of the instrument to distinguish different body components¹⁴ and serves as the gold standard for body composition measurements.¹⁵ With advances in technology, difficulties, such as anatomical level selection and imaging analysis of MRI instruments, have been overcome.¹⁶ Additionally, the absence of radiation exposure and reliable reproducibility makes MRI a promising technique for body composition assessments.¹⁷ Currently, reference ranges for body composition parameters assessed by MRI have not been reported; however, such studies using BIA and DXA have been published.^{18,19, S3} Hence, using data from the UK Biobank, we developed sex-specific reference ranges for body composition indices in a large community-dwelling population of middle-aged (aged 45–59 years) and young elderly (aged 60–74 years)

adults, hoping to supplement MRI-based body composition standards for future studies.

Methods

Participants

The UK Biobank, which was initiated in 2006, recruited more than half a million volunteers aged 40–69 years via the United Kingdom National Health Service registers between April 2007 and December 2010. After baseline assessment, an imaging visit was arranged for approximately 100 000 participants since 2014, covering brain, heart, and body composition MRI measurements.²⁰ The UK Biobank got ethical approval from the North West Multi-Center Research Ethics Committee (REC reference: 11/NW/03820), and all participants provided written informed consent. The corresponding research plan for this study had been submitted to the UK Biobank (Application ID 65814).

Our study screened participants with available MRI-measured body composition data from the UK Biobank cohort. Ten parameters were analysed, including three lean tissue parameters (total lean tissue volume, total thigh fat-free muscle volume [FFMV] and weight-to-muscle ratio) and seven adipose tissue parameters (muscle fat infiltration, abdominal subcutaneous adipose tissue [ASAT] volume, visceral adipose tissue [VAT] volume, total trunk fat volume, total adipose tissue volume, total abdominal adipose tissue index and abdominal fat ratio). Men and women aged 45–74 years with data on at least one of the parameters mentioned above were included. Ethnic minorities, defined as sample sizes, cannot support statistical estimation,²¹ and objective measurement mistakes based on documented error indicators were excluded from the analysis sets. We also excluded diseases, combined medications, surgeries and other physiological or pathological conditions that would influence the distribution of lean and fat mass, including endocrine, nutritional, and metabolic diseases (ICD-10: E00–07, E10–14, E15–16, E20–E32, E34–35, E40–46, E50–56, E58–61, E63, E64, E70–80, E83–90), neoplasms (ICD-10: C00–97, D00–09, D37–38), human immunodeficiency virus infection (ICD-10: B20–24), administration of anti-obesity preparations, systemic hormonal preparations and anti-diabetes products, pregnancy, and certain abdominal wall, subcutaneous tissue and miscellaneous surgeries (OPCS3: 400, 878–879, 928; OPCS4: S50–52, S62.2–62.9, S63.8–63.9, T32, T76–77, X02.1, X02.8–X02.9, X05, X09.1, X09.3, X09.8–09.9, X15, X22). A detailed participant's flowchart for each parameter is shown in File S1.

MRI protocol and body composition analysis

MRI was performed using a Siemens MAGNETOM Aera 1.5-T scanner (Siemens Healthineers, Erlangen, Germany) with the dual-echo Dixon Vibe protocol from the neck to knees. The imaging protocol provided a water- and fat-separated volumetric data set for body composition analysis. The detailed instrument process and analytical methods have been published elsewhere.^{22–24,S4} Briefly, the total 1.1-m scanning was divided into six overlapping slabs, with 64 slices in Slabs 1 and 6, 44 slices in Slabs 2 to 4 and 72 slices in Slab 5. The images obtained were calibrated, stack fused and segmented. After professional inspection by an analysis engineer, the lean and adipose volumes were determined.²⁵ The body composition analyses were performed using AMRA Profiler™ (AMRA AB, Linköping, Sweden).

Statistical analysis

The basic characteristics and body composition parameters of the enrolled participants were presented as mean (standard deviation) or median (interquartile range) based on data features. Considering that body composition varies between two sexes, we generated sex-specific percentile curves and age-and-sex-specific reference ranges using the lambda-mu-sigma (LMS) method.²⁶ The LMS method assumes underlying frequency distributions of the data and estimates each of the distribution moments in the form of smooth curves plotted against age. Box-Cox Cole and Green (BCCG), Box-Cox power exponential (BCPE) and Box-Cox t (BCT) are the LMS classes of distributions. The BCCG model estimates the first three moments of the measurement distribution as the age-varying median (μ), coefficient of variation (σ) and skewness in the form of a Box-Cox transformation (λ). The BCPE and BCT models extend the BCCG model by adjusting for kurtosis (τ) based respectively on the power exponential distribution and the t distribution. In this study, for each body composition parameter, we fitted three models and selected the optimal one based on the local maximum likelihood. We generated reference curves and provided reference ranges for both sexes at specific ages according to the estimated coefficients in the optimal model. The R package 'gamlss' was used for this analysis.

Additionally, according to the age segmentation criteria,²⁷ we divided the entire population into two subcategories: middle-aged (aged 45–59 years) and young elderly (aged 60–74 years) adults. We then conducted a consistency test (via Student's *t* test) to determine whether the measurement values in the subcategories had significant differences and thus to determine whether the reference ranges should be presented separately or jointly. The reference ranges were

presented as the mean $\pm$ 1 standard deviation (SD), mean $\pm$ 2 SDs, and mean $\pm$ 2.5 SDs.

Results

Participants' characteristics

The participants' characteristics are shown in *Table 1*. The number of participants varied for 10 parameters; hence, the data analysis sets of body composition were also different accordingly. The participant flowchart is shown in *Figure S1*. Because of the low statistical power of the ethnic minority, only participants with European ancestry were included in further analyses. After confirmation of the prescribed eligibility criteria, the final data sets for the parameters ranged from 4842 to 14 148.

For the three lean volume parameters, the median total lean tissue volume and total thigh FFMV were higher, and the median weight-to-muscle ratio was lower, in men than in women. Among the seven fat volume parameters, median muscle fat infiltration, ASAT volume, total adipose tissue volume, total abdominal adipose tissue index and abdominal fat ratio were higher in women than in men, but median VAT volume was higher in men than in women. Meanwhile, the remaining total trunk fat volume was comparable between women and men.

Lean volume parameters

The reference percentiles of the lean tissue parameters are listed in *Table 2*. Both total lean tissue volume and total thigh FFMV were progressively lower in older individuals of both sexes (*Figure 1*). In men and women, the median total lean tissue volume and total thigh FFMV were 2.83 and 1.73 L, and 3.02 and 1.51 L higher at age 45 years than at age 70 years, respectively. The weight-to-muscle ratio was higher in older individuals of both sexes (*Figure 1*). Compared with those at age 45, the median percentiles of weight-to-muscle ratio at age 70 were 0.51 and 0.83 kg/L higher in men and women, respectively.

Reference data on the lean volume parameters for middle-aged and young elderly adults are presented in *Table 3*. Three levels of ranges were given, which were equivalent to the mean $\pm$ 1 standard deviation (SD), mean $\pm$ 2 SDs, and mean $\pm$ 2.5 SDs. Consistency tests between the two populations were conducted in both sexes, and significant differences were found in each parameter (*Table S1*), indicating that the reference ranges for middle-aged and young adults should be independent from each other.

Table 1 Participants' characteristics

Parameter	Gender	Median (IQR)	Age (mean $\pm$ SD, years)	Height (mean $\pm$ SD, m)	Weight (median (IQR), kg)	BMI (median (IQR), kg/m ²)
Total lean tissue volume (L)	Male (n = 2122) Female (n = 2720)	28.37 (26.28, 30.67) 20.51 (18.94, 22.26)	60.63 $\pm$ 7.21 60.23 $\pm$ 7.10	1.76 $\pm$ 0.06 1.63 $\pm$ 0.06	81.20 (73.50, 89.00) 66.50 (59.90, 75.00)	26.13 (24.03, 28.45) 25.00 (22.61, 28.08)
Total thigh FFMV (L)	Male (n = 5697) Female (n = 7416)	12.41 (11.33, 13.55) 8.19 (7.45, 8.96)	61.28 $\pm$ 7.06 60.92 $\pm$ 6.86	1.76 $\pm$ 0.06 1.63 $\pm$ 0.06	80.80 (73.50, 89.20) 66.50 (59.60, 74.90)	26.04 (23.92, 28.51) 24.95 (22.55, 27.99)
Weight-to-muscle ratio (kg/L)	Male (n = 5697) Female (n = 7415)	6.49 (6.05, 7.03) 8.13 (7.44, 8.93)	61.28 $\pm$ 7.06 60.93 $\pm$ 6.86	1.76 $\pm$ 0.06 1.63 $\pm$ 0.06	80.80 (73.50, 89.20) 66.50 (59.60, 74.90)	26.04 (23.92, 28.51) 24.95 (22.55, 28.00)
Muscle fat infiltration (%)	Male (n = 5674) Female (n = 7388)	6.15 (5.33, 7.20) 7.30 (6.37, 8.44)	61.28 $\pm$ 7.05 60.91 $\pm$ 6.86	1.76 $\pm$ 0.06 1.63 $\pm$ 0.06	80.80 (73.50, 89.20) 66.50 (59.60, 74.90)	26.05 (23.92, 28.52) 24.95 (22.54, 27.98)
ASAT volume (L)	Male (n = 6544) Female (n = 7493)	5.23 (4.05, 6.74) 7.22 (5.37, 9.50)	61.12 $\pm$ 7.06 60.94 $\pm$ 6.86	1.77 $\pm$ 0.07 1.63 $\pm$ 0.06	81.80 (74.40, 90.60) 66.50 (59.60, 74.80)	26.20 (24.05, 28.73) 24.94 (22.54, 27.96)
VAT volume(L)	Male (n = 6574) Female (n = 7574)	4.39 (2.92, 6.00) 2.20 (1.39, 3.25)	61.12 $\pm$ 7.06 60.93 $\pm$ 6.87	1.77 $\pm$ 0.07 1.63 $\pm$ 0.06	81.90 (74.40, 90.80) 66.60 (59.70, 75.05)	26.22 (24.07, 28.76) 24.97 (22.58, 28.09)
Total abdominal adipose tissue index (L/m ²)	Male (n = 6539) Female (n = 7487)	3.13 (2.31, 4.03) 3.59 (2.59, 4.78)	61.11 $\pm$ 7.06 60.94 $\pm$ 6.87	1.77 $\pm$ 0.07 1.63 $\pm$ 0.06	81.80 (74.40, 90.60) 66.50 (59.60, 74.80)	26.20 (24.05, 28.73) 24.94 (22.54, 27.96)
Total adipose tissue volume (L)	Male (n = 2122) Female (n = 2720)	18.10 (14.48, 22.25) 20.30 (16.18, 25.37)	60.63 $\pm$ 7.21 60.23 $\pm$ 7.10	1.76 $\pm$ 0.06 1.63 $\pm$ 0.06	81.20 (73.50, 89.00) 66.50 (59.90, 75.00)	26.13 (24.03, 28.45) 25.00 (22.61, 28.08)
Total trunk fat volume (L)	Male (n = 5668) Female (n = 7329)	9.58 (7.09, 12.31) 9.53 (6.90, 12.62)	61.28 $\pm$ 7.05 60.92 $\pm$ 6.86	1.76 $\pm$ 0.06 1.63 $\pm$ 0.06	80.80 (73.50, 89.10) 66.40 (59.60, 74.60)	26.03 (23.92, 28.50) 24.92 (22.52, 27.89)
Abdominal fat ratio	Male (n = 5668) Female (n = 7329)	0.44 (0.37, 0.50) 0.54 (0.46, 0.61)	61.28 $\pm$ 7.05 60.92 $\pm$ 6.86	1.76 $\pm$ 0.06 1.63 $\pm$ 0.06	80.80 (73.50, 89.10) 66.40 (59.60, 74.60)	26.03 (23.92, 28.50) 24.92 (22.52, 27.89)

Abbreviations: ASAT, abdominal subcutaneous adipose tissue; FFMV, fat-free muscle volume; IQR, interquartile range; SD, standard deviation; VAT, visceral adipose tissue.

The reference values used in the LMS method for the fat volume parameters are available in *Data S1*.

Fat volume parameters

The reference percentiles of adipose tissue parameters are shown in *Table 4*. In men, four (muscle fat infiltration, VAT volume, total abdominal adipose tissue index and abdominal fat ratio) and two (ASAT volume and total adipose tissue volume) parameters were slightly higher and lower in older participants, respectively, and one parameter (total trunk fat volume) showed no significant difference (*Figures 2* and *S1*). Compared with those at age 45, the median muscle fat infiltration, VAT volume, total abdominal adipose tissue index and abdominal fat ratio were higher by 1.48%, 0.32 L, 0.08 L/m² and 0.04, respectively, in men at age 70. Meanwhile, the median ASAT volume and total adipose tissue volume were lower by 0.47 and 0.41 L, respectively, in men at age 70. In both groups, the median total trunk fat volume was approximately 9.53 L. In women, only muscle fat infiltration and VAT volume were similar to those in men (*Figures 3* and *S1*). The median percentiles of muscle fat infiltration were 6.25% and 7.93%, and the median VAT volumes were 1.61 and 2.37 L at ages 45 and 70, respectively. Total abdominal adipose tissue index and abdominal fat ratio were higher at age 60 than at 45 (3.65 L/m² vs. 3.16 L/m², and 0.55 vs. 0.49, respectively); the values at age 70 were 3.64 L/m² and 0.55, which were similar to those at age 60. The ASAT volume, total adipose tissue volume and total trunk fat volume were slightly higher at age 60 than at age 45 and were slightly lower at age 70 than at age 60 (*Figures 3* and *S1*). Compared with those at age 45 years, ASAT volume, total adipose tissue volume and total trunk fat volume were higher by 0.35, 0.78 and 1.12 L at age 60, respectively. Compared with those at age 60 years, they were lower by 0.33, 0.14 and 0.20 L at age 70, respectively.

Three levels of reference ranges for fat volume parameters are shown in *Table 5*, which were also equivalent to the mean $\pm$ 1 SD, mean $\pm$ 2 SDs and mean $\pm$ 2.5 SDs. Notably, the lower limits of certain parameters were below zero owing to the modelling computation. The reference ranges for these parameters should be treated unilaterally in consideration of clinical practice. Student's *t* tests between the two populations showed inconsistent results among the seven fat volume parameters (*Table S2*). The reference ranges for total trunk fat volume, total adipose tissue volume, total abdominal adipose tissue index in men and ASAT volume in women may be combined between middle-aged and young elderly adults. Nevertheless, based on population integrity and clinical application, we preferred the separation of reference data according to the age segmentation criteria.

Table 2 Reference percentiles of lean tissue parameters for both genders at specific ages

	Age (years)	Total lean tissue volume (L)	Total thigh FFMV (L)	Weight-to-muscle ratio (kg/L)
Male	45	30.14 (24.52, 38.19)	13.53 (10.34, 17.27)	6.19 (5.04, 7.84)
	50	29.62 (23.78, 36.75)	13.23 (10.14, 16.80)	6.28 (5.14, 7.91)
	55	29.07 (23.09, 35.60)	12.91 (9.93, 16.31)	6.37 (5.23, 8.01)
	60	28.50 (22.62, 34.73)	12.57 (9.69, 15.79)	6.47 (5.33, 8.12)
	65	27.91 (22.22, 33.94)	12.19 (9.42, 15.24)	6.58 (5.43, 8.27)
	70	27.31 (21.69, 33.01)	11.80 (9.14, 14.69)	6.70 (5.53, 8.44)
Female	45	22.75 (18.26, 28.20)	9.32 (7.13, 12.22)	7.47 (5.79, 9.85)
	50	21.78 (17.19, 27.43)	8.83 (6.80, 11.43)	7.77 (6.05, 10.27)
	55	20.97 (16.57, 26.41)	8.44 (6.53, 10.81)	8.04 (6.26, 10.66)
	60	20.45 (16.43, 25.33)	8.19 (6.35, 10.39)	8.18 (6.39, 10.87)
	65	20.06 (16.27, 24.64)	8.00 (6.21, 10.10)	8.23 (6.48, 10.88)
	70	19.73 (15.96, 24.29)	7.81 (6.05, 9.84)	8.30 (6.58, 10.91)

Note: Reference percentile was defined as median (3rd, 97th).

Abbreviation: FFMV, fat-free muscle volume.

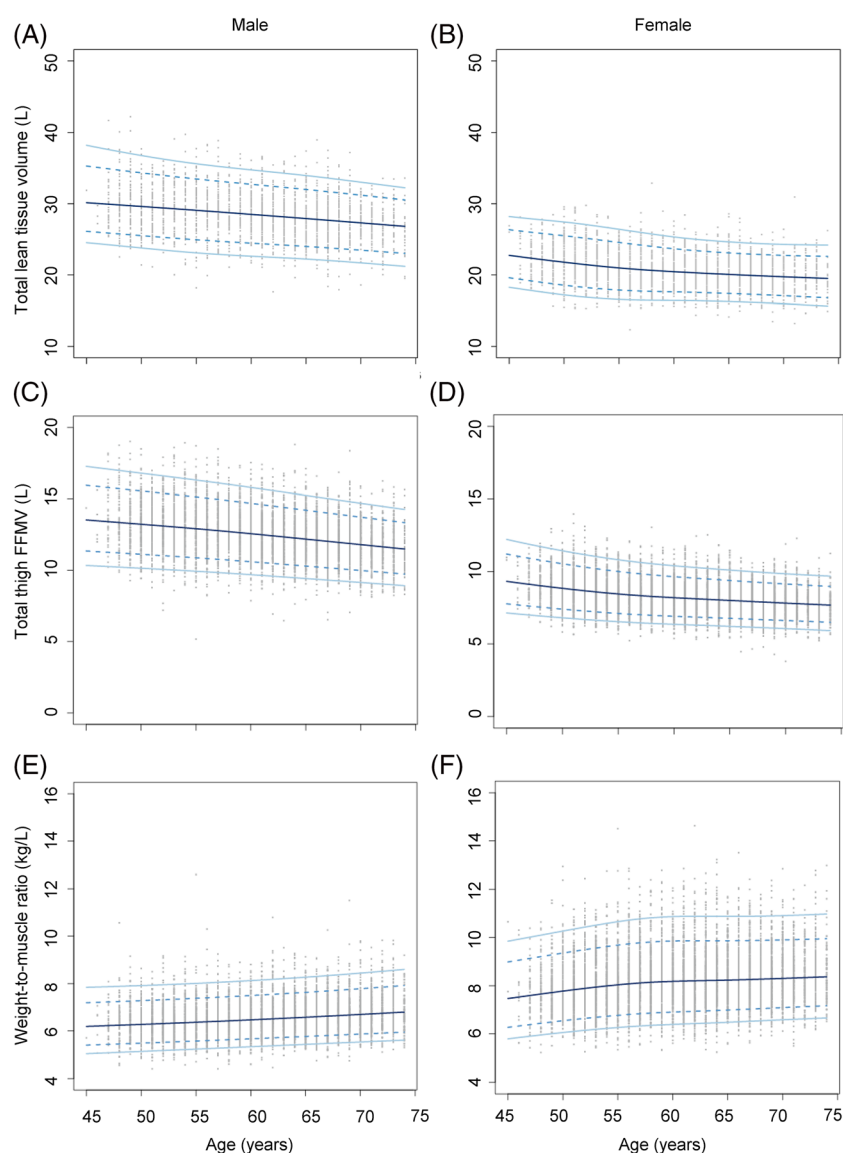


Figure 1 Age-specific and sex-specific percentile curves for (A) Total lean tissue volume in male, (B) total lean tissue volume in female, (C) total thigh fat-free muscle volume (FFMV) in male, (D) total thigh FFMV in female, (E) weight-to-muscle ratio in male and (F) weight-to-muscle ration in female

Table 3 Reference data of middle-aged adults and young elders for lean tissue parameters

	Parameters	Male			Female		
		Mean ± SD	Mean ± 2 SD	Mean ± 2.5 SD	Mean ± SD	Mean ± 2 SD	Mean ± 2.5 SD
Middle-aged adults (45–59 years)	Total lean tissue volume (L)	[25.98, 32.83]	[22.55, 36.26]	[20.83, 37.98]	[18.65, 24.09]	[15.92, 26.82]	[14.56, 28.18]
	Total thigh FFMV (L)	[11.33, 14.76]	[9.61, 16.48]	[8.75, 17.34]	[7.40, 9.78]	[6.21, 10.97]	[5.62, 11.57]
	Weight-to-muscle ratio (kg/L)	[5.67, 7.17]	[4.92, 7.92]	[4.54, 8.29]	[6.91, 9.27]	[5.73, 10.45]	[5.14, 11.04]
Young elders (60–74 years)	Total lean tissue volume (L)	[24.71, 30.93]	[21.60, 34.04]	[20.04, 35.60]	[17.90, 22.40]	[15.65, 24.65]	[14.53, 25.78]
	Total thigh FFMV (L)	[10.57, 13.68]	[9.01, 15.24]	[8.23, 16.02]	[6.97, 9.06]	[5.93, 10.10]	[5.40, 10.63]
	Weight-to-muscle ratio (kg/L)	[5.92, 7.46]	[5.15, 8.23]	[4.77, 8.61]	[7.17, 9.58]	[5.97, 10.78]	[5.37, 11.38]

Abbreviations: FFMV, fat-free muscle volume; SD, standard deviation.

Table 4 Reference percentiles of fat tissue parameters for both genders at specific ages

Age (years)	Muscle fat infiltration (%)	Total abdominal adipose tissue index (L/m ²)			Total adipose tissue volume (L)	Total trunk fat volume (L)	Abdominal fat ratio
		ASAT volume (L)	VAT volume (L)	adipose tissue index (L/m ²)			
Male	45	5.25 (3.74, 8.09)	4.09 (0.96, 9.25)	3.06 (0.98, 6.38)	18.41 (7.20, 33.07)	9.52 (3.27, 19.59)	0.41 (0.19, 0.58)
	50	5.49 (3.91, 8.49)	4.17 (1.04, 9.25)	3.08 (1.05, 6.30)	18.31 (7.83, 32.61)	9.52 (3.35, 19.30)	0.42 (0.20, 0.59)
	55	5.76 (4.09, 8.93)	4.24 (1.12, 9.25)	3.10 (1.11, 6.22)	18.22 (8.34, 32.20)	9.53 (3.43, 19.02)	0.43 (0.21, 0.59)
	60	6.05 (4.29, 9.42)	4.30 (1.19, 9.26)	3.11 (1.15, 6.15)	18.14 (8.76, 31.82)	9.53 (3.51, 18.74)	0.43 (0.22, 0.59)
	65	6.38 (4.51, 9.96)	4.36 (1.25, 9.28)	3.12 (1.19, 6.06)	18.06 (9.11, 31.47)	9.53 (3.60, 18.45)	0.44 (0.23, 0.60)
	70	6.73 (4.75, 10.56)	4.41 (1.30, 9.31)	3.14 (1.20, 5.96)	18.00 (9.41, 31.15)	9.53 (3.69, 18.17)	0.45 (0.24, 0.60)
	45	6.25 (4.08, 10.09)	1.61 (0.43, 5.06)	3.16 (1.10, 7.19)	19.70 (9.43, 38.92)	8.60 (2.93, 20.28)	0.49 (0.25, 0.66)
Female	50	6.57 (4.58, 9.86)	1.85 (0.49, 5.35)	3.33 (1.18, 7.32)	20.09 (9.81, 38.53)	9.02 (3.14, 20.14)	0.51 (0.27, 0.67)
	55	6.90 (4.88, 10.25)	2.08 (0.55, 5.63)	3.53 (1.26, 7.52)	20.39 (10.14, 38.13)	9.47 (3.36, 20.22)	0.53 (0.29, 0.69)
	60	7.24 (5.03, 11.10)	2.24 (0.60, 5.73)	3.65 (1.32, 7.58)	20.48 (10.35, 37.47)	9.72 (3.52, 19.99)	0.55 (0.30, 0.70)
	65	7.58 (5.38, 11.40)	2.32 (0.62, 5.67)	3.66 (1.34, 7.39)	20.41 (10.47, 36.61)	9.65 (3.56, 19.21)	0.55 (0.31, 0.69)
	70	7.93 (5.61, 12.10)	2.37 (0.63, 5.57)	3.64 (1.36, 7.16)	20.34 (10.58, 35.85)	9.52 (3.58, 18.42)	0.55 (0.33, 0.69)

Note: Reference percentile was defined as median (3rd, 97th).
Abbreviations: ASAT, abdominal subcutaneous adipose tissue; VAT, visceral adipose tissue.

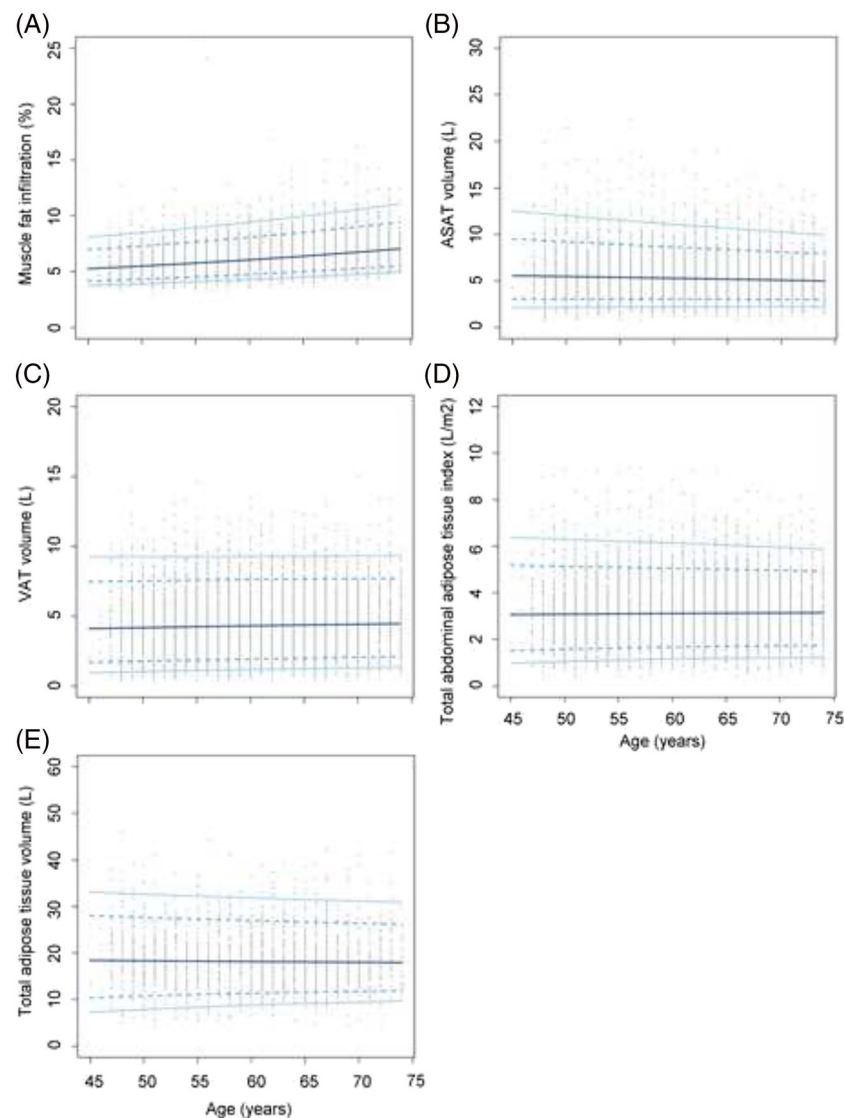


Figure 2 Age-specific and sex-specific percentile curves for (A) muscle fat infiltration, (B) abdominal subcutaneous adipose tissue (ASAT) volume, (C) visceral adipose tissue (VAT) volume, (D) total abdominal adipose tissue index and (E) total adipose tissue volume in male.

The reference values used in the LMS method for the fat volume parameters are available in *Data S1*.

Discussion

Using MRI data from the UK Biobank, we established sex-specific reference ranges of abdominal body composition for middle-aged and young elderly adults in a natural population. To our knowledge, this is the first study to develop reference ranges for MRI-measured body composition parameters using a large sample size. As the gold standard for body composition assessment, the reference data generated from MRI in this study can help researchers develop more ac-

curate diagnostic criteria for overweight/obesity, sarcopenia, amyotrophy and other metabolic disorders. The introduction of age segmentation makes reference ranges more practical and accurate in clinical practice. Additionally, the established reference ranges may also benefit from the renovation of classic disease risk prediction models to achieve better efficiency.

Recent studies have reported the reference ranges of body composition parameters evaluated by BIA and DXA in various countries and regions.^{19,55-56} Because of the different assessment tools and measurement units, the results of this study cannot be directly compared with previous findings; however, age-related relationships of similar parameters can be roughly compared. For lean parameters, our results were

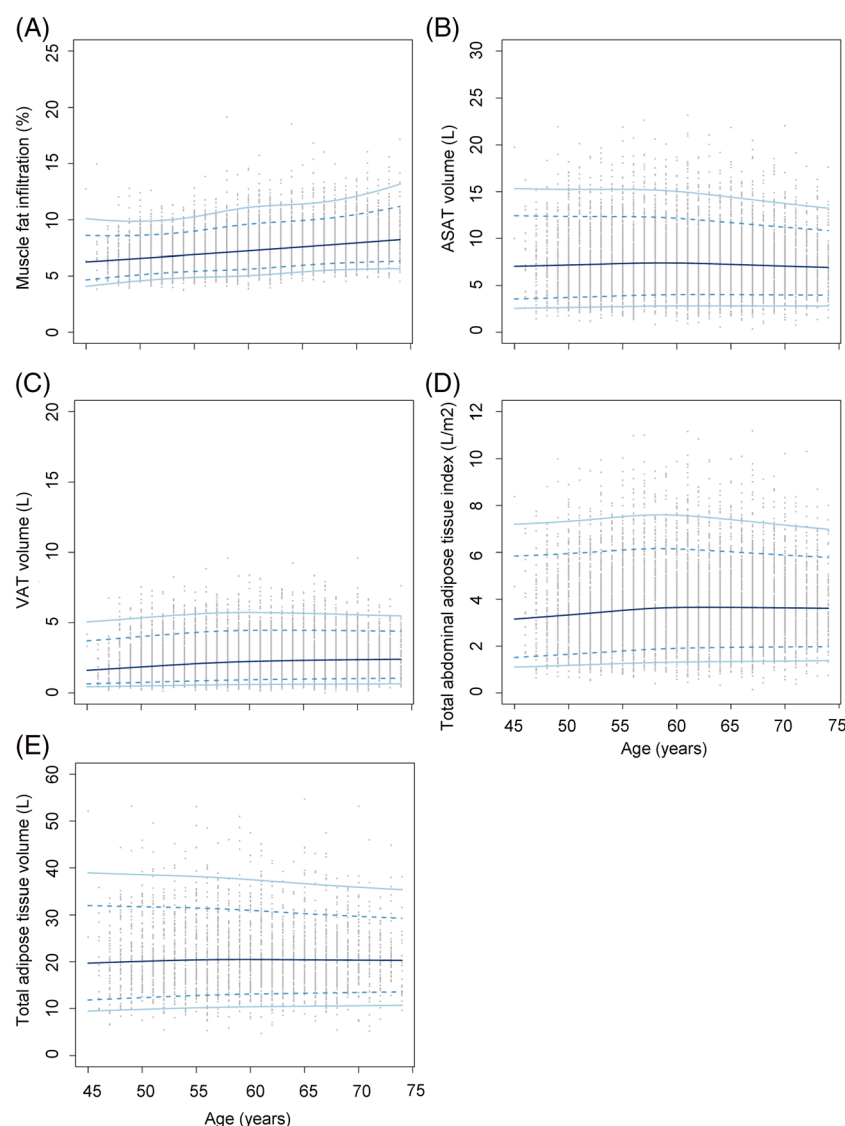


Figure 3 Age-specific and sex-specific percentile curves for (A) muscle fat infiltration, (B) abdominal subcutaneous adipose tissue (ASAT) volume, (C) visceral adipose tissue (VAT) volume, (D) total abdominal adipose tissue index and (E) total adipose tissue volume in female.

consistent with those of previous studies in Caucasians^{28,29}; specifically lean tissue contents were lower at older ages. Interestingly, the fat volume parameters in our study did not follow those reported in recent studies. Although VAT volume and muscle fat infiltration were higher at older ages, other parameters were either similar or lower at 70 years old. According to a previous study using UK Biobank data,³⁰ the BIA-assessed fat mass index showed a gradually increasing trend with ageing, which we speculate may have resulted from impedance changes in adipose tissue⁵⁷. It has been well-acknowledged that human adipose tissue is composed of white adipose tissue (WAT) and brown adipose tissue (BAT).^{31,58} WAT adipocyte contains large lipid droplet, while BAT adipocytes are characterized by abundant mitochondria⁵⁹. That is, based on BIA rationale⁵¹⁰, WAT has

lower impedance and higher mass compared to BAT. Despite the total adipose tissue volume may initially be balanced, studies have reported that the contents of subcategories and fat distribution are constantly changing.^{32,33} With age, both the number and volume of WAT adipocytes increase, whereas those of BAT decline. This finding may appropriately explain the paradox between our results and the previous report. Hence, we have reason to speculate that beneath the stable volume of adipose tissue, the composition of adipose tissue may have been altered; specifically, WAT increases while BAT decreases with ageing, which may further lead to an increase in adipose mass.

BMI and WC are fundamental parameters in diagnosing obesity. However, based on the definition of obesity, which is characterized by an increase in body fat stores,³⁴ simple an-

Table 5 Reference data of middle-aged adults and young elders for fat tissue parameters

Parameters	Male			Female		
	Mean $\pm$ SD	Mean $\pm$ 2 SD	Mean $\pm$ 2.5 SD	Mean $\pm$ SD	Mean $\pm$ 2 SD	Mean $\pm$ 2.5 SD
Middle-aged adults (45–59 years)						
Muscle fat infiltration (%)	[4.51, 7.30]	[3.12, 8.69]	[2.42, 9.38]	[5.53, 8.52]	[4.04, 10.02]	[3.29, 10.76]
ASAT volume (L)	[3.24, 8.34]	[0.69, 10.89]	[−0.58, 12.17]	[4.40, 11.15]	[1.02, 14.53]	[−0.67, 16.22]
VAT volume (L)	[2.27, 6.72]	[0.05, 8.94]	[−1.06, 10.06]	[0.92, 3.73]	[−0.49, 5.14]	[−1.19, 5.84]
Total abdominal adipose tissue index (L/m ²)	[1.86, 4.63]	[0.48, 6.01]	[−0.21, 6.70]	[2.03, 5.45]	[0.31, 7.16]	[−0.54, 8.02]
Young elders (60–74 years)						
Total adipose tissue volume (L)	[12.24, 25.29]	[5.71, 31.82]	[2.45, 35.08]	[13.77, 29.09]	[6.11, 36.75]	[2.28, 40.58]
Total trunk fat volume (L)	[5.76, 14.20]	[1.54, 18.42]	[−0.57, 20.53]	[5.49, 14.58]	[0.95, 19.13]	[−1.33, 21.40]
Abdominal fat ratio	[0.32, 0.52]	[0.22, 0.62]	[0.17, 0.67]	[0.40, 0.63]	[0.29, 0.74]	[0.24, 0.79]
Muscle fat infiltration (%)	[5.19, 8.25]	[3.66, 9.78]	[2.90, 10.55]	[6.17, 9.62]	[4.44, 11.35]	[3.58, 12.21]
ASAT volume (L)	[3.28, 7.74]	[1.06, 9.96]	[−0.05, 11.07]	[4.51, 10.66]	[1.44, 13.73]	[−0.10, 15.27]
VAT volume (L)	[2.45, 6.86]	[0.25, 9.07]	[−0.85, 10.17]	[1.17, 3.94]	[−0.21, 5.32]	[−0.90, 6.02]
Total abdominal adipose tissue index (L/m ²)	[1.96, 4.58]	[0.66, 5.88]	[0.01, 6.54]	[2.23, 5.46]	[0.62, 7.07]	[−0.19, 7.88]
Middle-aged adults (45–59 years)						
Total adipose tissue volume (L)	[12.75, 24.69]	[6.78, 30.66]	[3.79, 33.64]	[14.29, 28.11]	[7.37, 35.03]	[3.92, 38.48]
Total trunk fat volume (L)	[6.00, 13.94]	[2.03, 17.91]	[0.04, 19.89]	[5.92, 14.29]	[1.73, 18.47]	[−0.36, 20.57]
Abdominal fat ratio	[0.34, 0.53]	[0.24, 0.63]	[0.19, 0.68]	[0.43, 0.64]	[0.33, 0.74]	[0.28, 0.79]

Abbreviations: ASAT, abdominal subcutaneous adipose tissue; SD, standard deviation; VAT, visceral adipose tissue.

thropometric indices, such as BMI and WC, apparently do not suffice. Moreover, concepts, such as visceral obesity and localized adiposity, have garnered increased attention in recent years. Hence, a detailed assessment of adipose tissue distribution is necessary in clinical practice. DXA, which is a routine tool for quantifying body composition, can provide specific data on lean and fat volumes in the human body.³⁵ As a two-dimensional scanning technique, all imaging data were projected onto a single flat plane.²² Hence, DXA cannot distinguish between visceral, deep and superficial fat volumes. In contrast, MRI is a three-dimensional imaging technique. The introduction of the vertical axis can obtain body imaging information from different circumferential angles. Therefore, it can provide direct volumetric measurements of adipose tissue depots in different body parts.²² Nonetheless, medical expenditure for MRI is significantly higher than that for DXA. Therefore, clinicians should determine the most appropriate tool based on the medical and patient needs.

Sarcopenia, which is defined as age-related skeletal muscle loss, has received great attention in recent decades.^{36,511} Sarcopenia is not only related to various adverse health outcomes but also dramatically increases medical expenditure⁵¹². Due to the increasing prevalence of sarcopenia, a simple and accurate diagnostic method is particularly important for clinical practice. Currently, three parameters are evaluated when diagnosing sarcopenia: skeletal muscle strength, skeletal muscle mass and physical performance.^{15,37} Among these, muscle mass is the confirmatory index. Although MRI is the gold standard for noninvasive evaluation of muscle mass, it is not superior over BIA or DXA when diagnosing sarcopenia because cost-effectiveness is a high-priority consideration, and high precision is not necessary in routine practice. Therefore, although we provided reference ranges for MRI-based body composition parameters, assessing and diagnosing sarcopenia using MRI is currently considered aggressive.

In addition to routine screening, MRI is more suitable for refined evaluations of body composition. Muscle quality, which is a relatively new concept, indicates that changes in muscle architecture and composition would impact muscle function³⁸ and can be directly assessed using highly sensitive imaging instruments such as computed tomography and MRI⁵¹³. Skeletal muscle fat infiltration, also known as myosteatosis, is the primary indicator of muscle quality and recognized as an independent risk factor for muscle mass, strength and mobility after adjusting other body composition parameters.³⁹ Moreover, the accumulation of fat in muscle negatively impacts body metabolic status, and it may be further linked to an elevated risk of various health consequences such as diabetes, fractures, late recovery and mortality^{514–515}. Owing to the synergetic effect of sarcopenia and myosteatosis, it is increasingly recognized that muscle mass and quality should be evaluated simultaneously in determining risk stratification in the aging population. As a solution,

MRI is an appropriate evaluation method. We suggest that, if conditions permit, MRI should be performed for individuals who are at greater risk for sarcopenia and may desire detailed health instructions.

The findings of this study filled gap in the MRI-based body composition reference data. Relevant clinicians and researchers should apply these data wisely. Notably, the MRI scan covered only from neck (vertebrae T9) to knees (the bottom of the thigh muscles) of participants. The parameters regarding the total body tissues actually represent a knee-to-neck acquisition. The generalizability of the reference ranges should not be over-enlarged. Besides, three levels of ranges were calculated according to conventional rules. Considering the accuracy of MRI and the large sample size, we recommend more rigorous reference ranges. Future researchers may consider applying these reference values as grouping criteria and, then design longitudinal studies to explore the relationship between body composition and health-related outcomes. Once the accumulated data is sufficient, we can amend and finalize the reference ranges for body composition parameters.

Our study has certain inherent limitations that could influence the use of established reference ranges. Except for the limited generalizability of the total body tissues' parameters, all MRI-based body composition data were obtained from the UK Biobank cohort, which only included participants aged 45 years and older. The estimation of reference values for the entire population cannot be implemented, and further studies should be conducted to fill this gap. Additionally, owing to the limited sample size of ethnic minorities, the reference ranges may only be suitable for European Caucasians; hence, studies including ethnic minorities should be conducted in other regions. Furthermore, this study used a cross-sectional design; longitudinal data were not available. The age-based curves of body composition parameters were statistically simulated based on independent single-timepoint data; thus, they cannot be strictly defined as intra-individual trajectories, and the clinical application of these curves should be performed with caution. Moreover, the loss of longitudinal characteristics may further introduce a detection bias in our estimation. Currently, the UK Biobank conducts the repeated imaging tests. When data are sufficient, the reference ranges can be adjusted to obtain better accuracy.

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Conclusions

In summary, we established reference ranges for MRI-measured body composition parameters based on the UK Biobank study. The data would provide robust evidence for researchers and doctors to define abnormal adipose and muscle conditions in middle-aged and young elderly adults. Further studies are needed to supplement reference data on other age groups and ethnicities and intra-individual trajectories as well as to explore their relationship with health outcomes.

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The authors certify that they comply with the ethical guidelines for publishing in the *Journal of Cachexia, Sarcopenia and Muscle*: update 2021.⁴⁰

Conflict of interest

All authors declare no conflicts of interest regarding this study.

Online supplementary material

Additional supporting information may be found online in the Supporting Information section at the end of the article.

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